Candidate's Name: ……..

Signature: ……..

STREAM:							

525/1
CHEMISTRY
Paper 1
August
2025
$2\frac{1}{2}$ hours

END OF TERM II ASSESSMENT

Uganda Advanced Certificate of Education

S.5 CHEMISTRY

Paper 1

(*Set II*)

2 hours 30 minutes

INSTRUCTIONS TO CANDIDATES:

This paper has **two** *sections* **A** *and* **B.**

Section **A** *has two compulsory items while* **B** *has* **four** *items answer only* **three** *in* **B**.

Answers to Section **A** ***must*** *be written in the spaces provided and Section* **B** ***must*** *be written in the answer booklet(s) provided*

Answer five in all.

Where necessary use,

Molar gas volume at s.t.p = 22.4dm^3

FOR EXAMINER'S USE ONLY																													
ITEM	**1**							**2**					**3**				**4**				**5**				**6**				**TOTAL**
CODE																													
SCORE																													

SECTION A

Answer **all** *questions in this section*

ITEM 1

At a Kampala specialty chemicals firm, researchers have isolated a chlorinated organic compound, **Q**, from oilseed residues. Suspected to be a chlorinated hydrocarbon, **Q** is targeted as a precursor for antifungal agents for maize, a building block for aromatic solvents for a paint maker in Wakiso, and a starting monomer for plastic packaging.

In pre-tests, a 3.0 g sample of **Q** burnt in excess oxygen produced 2.67 g of CO_2 and 1.09 g of H_2O. The label indicates a vapour density of 49.5. When refluxed with excess alcoholic KOH, **Q** yielded a volatile product F that burned with a very sooty flame, raising concerns about safety.

The firm plans two product lines: converting **Q** into aromatic compound **R**, later derivatized into solvent $\mathbf{R_1}$ for paints, and into compound **D**, an alkene monomer for plastics. Compound **D** also reacts with HBr under different conditions to yield antifungal agents for maize. You have been contacted to assist.



https://share.google/images/yjDHu6ulKyOzjhLnj

Task

As a learner of Chemistry, guide the company on how to;

(a) Determine Molecular formula of compound **Q**.

(b) Show the analyst using an equation formation of compound **F**

(c) Predict the properties, possible environmental dangers and their mitigations that result from **F**

(d) Synthesis the solvent and monomer

(e) Explain mechanisms of reaction of monomer **D** with hydrogen bromide through two different ways yielding antifungal agents for post-harvest treatment of maize

(f) Evaluate the suitability of the aromatic compound **R** and its derivative $\mathbf{R_1}$ produced to be used in the industry

(g) Give the IUPAC names and structures of all the chemicals substances the company intends to use

ITEM 2

At the Uganda Nuclear Research Center, scientists are investigating two radioisotopes of importance in medicine and industry: Iodine-131 and Cesium-137. Both are widely applied in diagnostics and gauging, differing in half-life but alike in undergoing beta decay that releases significant nuclear energy. Students find it difficult to understand their atomic structures, decay behavior, and safety considerations, which are essential for practical use and environmental protection. To aid learning, activity (counts per hour) was measured over time (hours)

Time (hours)	4	6	8	10	12	14	16
Iodine-131 Activity (cps)	320	180	100	56	32	18	10
Cesium-137 Activity (cps)	600	425	300	210	150	105	75

The Principal has organized a workshop on these results, and you have been invited to give guidance.

Task

As the learner of chemistry help the students to;

(a) Come up with the electronic configurations and predict the locations in the periodic table and the nature of the isotopes given.

(b)(i) Explain the concept of radioactivity and the nuclear transformations, identifying isotopes formed for both substances above.

(ii) Analyze the data in the table in order to determine the decay constants and half-lives for both isotopes. Comment on your results for medical application.

(c) (i) Describe the practical applications of the process relevant to Uganda's sectors.

(d) (ii) Explain the possible side effects of the process and how they can be mitigated.

SECTION B

Attempt **three** *items in this section*

Item 3

A school chemistry club is investigating heat production from the formation of ionic compounds for use in small portable heaters during science exhibitions. They are considering three compounds: **X**, **Y**, and **Z**.

The club has collected thermochemical data in kJ mol^{-1} for the compounds are as follows;
For **X** (lithium fluoride): enthalpy of atomization of fluorine (+150), electron affinity of fluorine (–351), enthalpy of atomization of lithium (+155), first ionization energy of lithium (+518), and lattice energy (–1030).
For **Y** (calcium oxide): enthalpy of atomization of calcium (+117), first and second ionization energies of calcium (+59, +1100), heat of atomization of oxygen (+249), first and second electron affinities of oxygen (–141, +790), and lattice energy (–2810).
For **Z** (aluminium chloride): enthalpy of atomization of aluminium (+326), first, second, and third ionization energies of aluminium (+578, +1817, +2745), bond dissociation energy of chlorine (+242), electron affinity of chlorine (–364), and lattice energy (–5290).
All values are in kilojoules per mole.

The club wants to select the reaction that produces the most heat while remaining safe for a demonstration. You have been invited to provide guidance.

Task

As a student of Chemistry, Outline the method you would use to help the club choose between the three substances, and justify your choice based on your calculations and safety considerations.

ITEM 4

At the National Agricultural Research Organisation in Arua district, soil within the area was found to contain a sodium salt, $NaHCO_3 \cdot XH_2O$, affecting crop growth.

In analysis, 6.0 g of **$NaHCO_3 \cdot xH_2O$** was dissolved to one liter, and 25cm^3 of this solution required 15cm^3 of 0.05M hydrochloric acid for complete neutralization.

The organization seeks to determine the actual formula of the salt, its nature, and soil pH, and to make recommendations for improving crop yields. You have been contacted to provide guidance.

Task

As a chemistry student, make a write up you will use to help the organization.

ITEM 5

An environmental agency is investigating air pollution near an industrial zone suspected of hydrocarbon release and open combustion. A 20.0 cm^3 sample of hydrocarbon **Z** was

reacted with 200.0 cm^3 of oxygen; after cooling, total gas volume was 160.0 cm^3, and passing residual gases through aqueous sodium hydroxide reduced the volume to 20.0 cm^3.
Hydrocarbon **Z** burns with a sooty flame. When treated with hot alkaline potassium manganate(VII) and then acidified with dilute hydrochloric acid, it produces compound **T**, which reacts with magnesium to liberate hydrogen.

The agency seeks to understand the chemical equation for combustion of **Z**, the molecular formula, IUPAC names and structures of **Z** and **T**, a method to synthesize **Z** from ethanol, and the environmental hazards of **Z** with mitigation measures. You have been contacted to provide guidance.

Task

As a student of Chemistry, prepare a presentation you will use upon invitation.

ITEM 6

A scientist is studying the Periodic properties of several elements and needs to determine the trend of atomic radius across the period and down the group in the Periodic Table. He is focusing on the elements in the third period from sodium to chlorine and elements in group (II) from beryllium to barium. He obtained the following data but lacks explanation for the two trends.

Element	**Be**	**Na**	**Mg**	**Al**	**Si**	**P**	**S**	**Cl**	**Ca**	**Sr**	**Ba**
Atomic number	4	11	12	13	14	15	16	17	20	38	56
Atomic radius (nm)	0.089	0.156	0.136	0.125	0.117	0.110	0.104	0.099	0.174	1.191	0.198

The scientist seeks guidance to interpret the data for Period 3 and Group II elements and to analyze and explain the observed trends in atomic radius for use in a science exhibition presentation.. You have approached for guidance.

Task

As a learner of chemistry prepare a write up to help him.

THE PERIODIC TABLE

1	2											3	4	5	6	7	8
1.0 H 1																1.0 H 1	4.0 He 2
6.9 Li 3	9.0 Be 4											10.8 B 5	12.0 C 6	14.0 N 7	16.0 O 8	19.0 F 9	20.2 Ne 10
23.0 Na 11	24.3 Mg 12											27.0 Al 13	28.1 Si 14	31.0 P 15	32.1 S 16	35.4 Cl 17	40.0 Ar 18
39.1 K 19	40.1 Ca 20	45.0 Sc 21	47.9 Ti 22	50.9 V 23	52.0 Cr 24	54.9 Mn 25	55.8 Fe 26	58.9 Co 27	58.7 Ni 28	63.5 Cu 29	65.7 Zn 30	69.7 Ga 31	72.6 Ge 32	74.9 As 33	79.0 Se 34	79.9 Br 35	83.8 Kr 36
85.5 Rb 37	87.6 Sr 38	88.9 Y 39	91.2 Zr 40	92.9 Nb 41	95.9 Mo 42	98.9 Tc 43	101 Ru 44	103 Rh 45	106 Pd 46	108 Ag 47	112 Cd 48	115 In 49	119 Sn 50	122 Sb 51	128 Te 52	127 I 53	131 Xe 54
133 Cs 55	137 Ba 56	139 La 57	178 Hf 72	181 Ta 73	184 W 74	186 Re 75	190 Os 76	192 Ir 77	195 Pt 78	197 Au 79	201 Hg 80	204 Tl 81	207 Pb 82	209 Bi 83	209 Po 84	210 At 85	222 Rn 86
223 Fr 87	226 Ra 88	227 Ac 89															

		139 La 57	140 Ce 58	141 Pr 59	144 Nd 60	147 Pm 61	150 Sm 62	152 Eu 63	157 Gd 64	159 Tb 65	162 Dy 66	165 Ho 67	167 Er 68	169 Tm 69	173 Yb 70	175 Lu 71
		227 Ac 89	232 Th 90	231 Pa 91	238 U 92	237 Np 93	244 Pu 94	243 Am 95	247 Cm 96	247 Bk 97	251 Cf 98	254 Es 99	257 Fm 100	256 Md 101	254 No 102	260 Lw 103

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